



Computer Vision

L04: Feature Detection and Matching (Scale Invariant Feature Transform (SIFT))

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Outline

What IS SIFT?

Steps for Extracting Key Points

- Scale Space Peak Selection

- Key Point Localization

- Orientation Assignment

- Key Point Descriptor

Key Point Matching



Outline

What IS SIFT?

Steps for Extracting Key Points

Key Point Matching



SIFT - Key Point Extraction

- ▶ Stands for **Scale Invariant Feature Transform**.
- ▶ Similar to the one used in **primate visual system** (human, ape, monkey, etc.).
- ▶ Transforms image data into **scale-invariant coordinates**.
- ▶ **Histograms of oriented gradients**

Goal

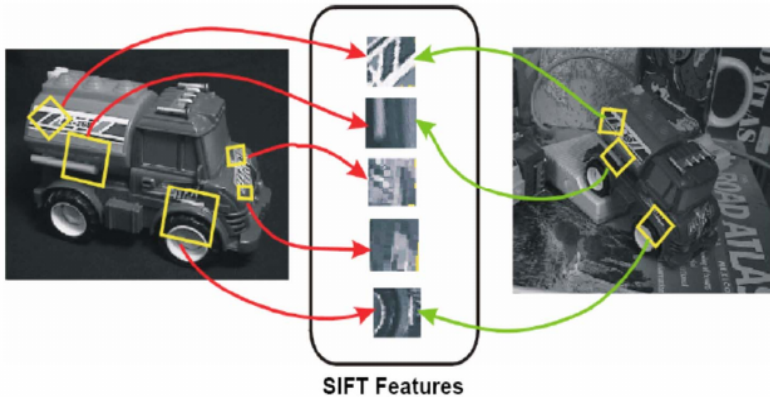
- ▶ Extracting **distinctive invariant features**
- ▶ **Correctly matched against a large database** of features from many images.
- ▶ **Invariance** to image **scale** and **rotation**.
- ▶ **Robustness** to
 - ▶ Affine distortion: Examples of affine transformations include translation, scaling, homothety, similarity transformation, reflection, and rotation.
 - ▶ Change in 3D viewpoint.
 - ▶ Addition of noise.
 - ▶ Change in illumination.



Advantages

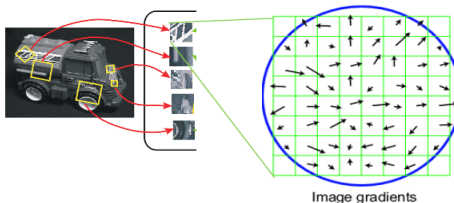
- ▶ **Locality**: features are local, so robust to occlusion and clutter.
- ▶ **Distinctiveness**: individual features can be matched to a large database of objects.
- ▶ **Quantity**: many features can be generated for even small objects.
- ▶ **Efficiency**: close to real-time performance.

Invariant Local Features



SIFT Vector Formation

- ▶ Computed on rotated and scaled version of window according to computed orientation and scale $\Rightarrow$ resample the window.
- ▶ Based on gradients weighted by a Gaussian of variance half the window (for smooth falloff).





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Steps for Extracting Key Points

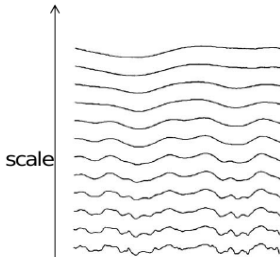
- ▶ **Scale space peak selection:** Potential locations for finding features.
- ▶ **Key point localization:** Accurately locating the feature key points.
- ▶ **Orientation Assignment:** Assigning orientation to the key points.
- ▶ **Key point descriptor:** Describing the key point as a high dimensional vector.

Scales

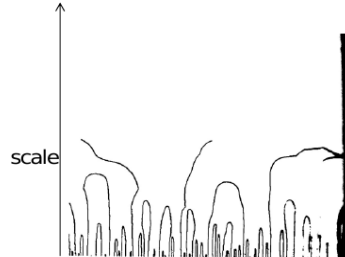
- ▶ What should be **sigma value** for **Canny** and **LoG edge detection**?
- ▶ If use multiple sigma values (scales), how do you **combine multiple edge maps**?
- ▶ **Marr-Hildreth** $\Rightarrow$ **Spatial Coincidence assumption**:
Zero crossings that coincide over several scales are physically significant.
 - ▶ Does not really correct and does not work.

Scale Space

- ▶ Apply **whole spectrum of scales**.
- ▶ Plot **zerocrossings** vs **scales** in a scale-space.

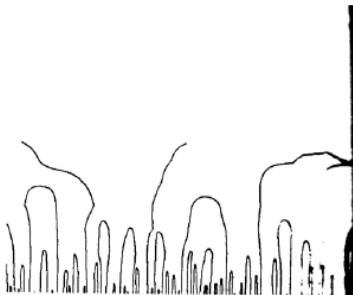


Multiple smooth versions of a signal

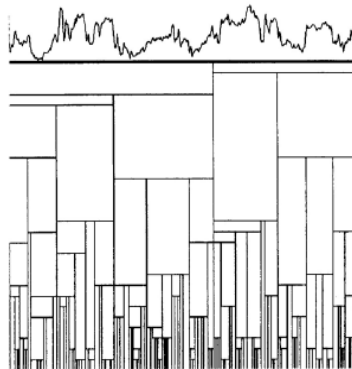


Zerocrossings at multiple scale

Scale Space (cont'd)



Scale Space



Interval Tree

Scale Space (cont'd)

- ▶ Apply **whole** spectrum of scales.
- ▶ Plot **zerocrossings** vs **scales** in a scale-space.
- ▶ **Interpret scale space contours:**
 - ▶ Contours are arches, open at the bottom, closed at the top.
 - ▶ **Interval tree:**
 - ▶ Each interval corresponds to a node in a tree, whose parent node represents larger interval, from which interval emerged, and whose off springs represent smaller intervals.
 - ▶ Stability of a node is a scale range over which the interval exists.



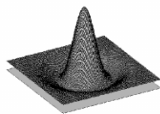
Scale Space (cont'd)

- ▶ **Top level description:**
 - ▶ Iteratively remove nodes from the tree, splicing out nodes that are less stable than any of their parents and off springs.

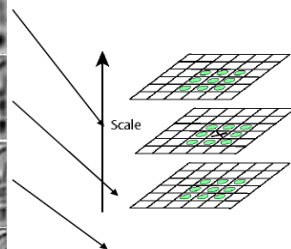
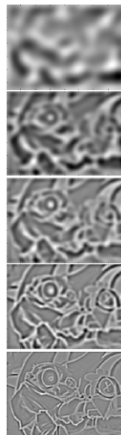
Laplacian-of-Gaussian (LoG)

Interest points:

Local maxima in scale space of Laplacian-of-Gaussian



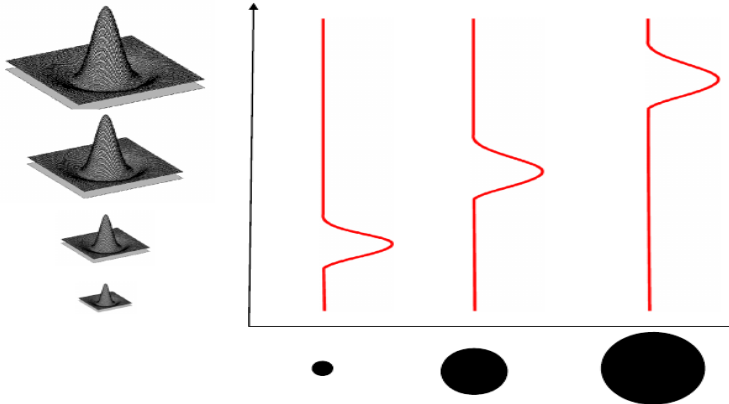
$$L_{xx}(\sigma) + L_{yy}(\sigma) \rightarrow \begin{matrix} \nearrow \sigma^5 \\ \nearrow \sigma^4 \\ \rightarrow \sigma^3 \\ \searrow \sigma^2 \\ \searrow \sigma \end{matrix}$$



$\Rightarrow$ **List of**
(x, y, σ)

What Is A Useful Signature Function?

Laplacian-of-Gaussian = “blob” detector



Scale-space blob detector: Example



Scale-space blob detector: Example ($\sigma=2$)



Scale-space blob detector: Example ($\sigma=2.5018$)



Scale-space blob detector: Example ($\sigma=3.1296$)



Scale-space blob detector: Example ($\sigma=3.9149$)



Scale-space blob detector: Example ($\sigma=4.5972$)



Scale-space blob detector: Example ($\sigma=5.126$)



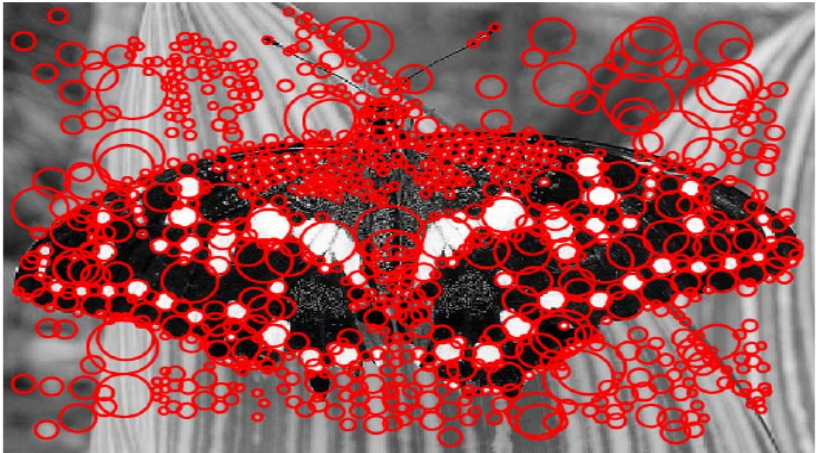
Scale-space blob detector: Example ($\sigma=9.5859$)



Scale-space blob detector: Example ($\sigma=11.9912$)

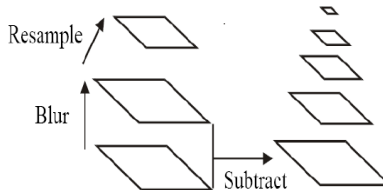


Scale-space blob detector: Example (Result)



Building a Scale Space

- ▶ All scales must be examined to identify scale-invariant features.
- ▶ An efficient function is to compute the Laplacian Pyramid (Difference of Gaussian).



Approximation of LoG by Difference of Gaussians

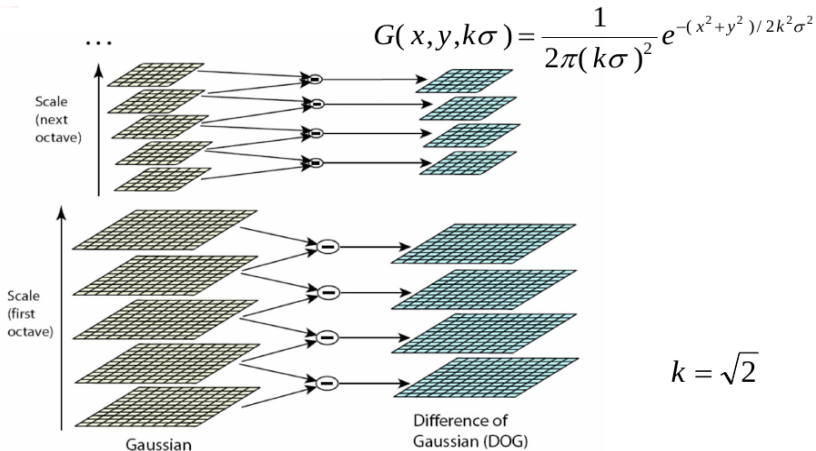
$$\frac{\partial G}{\partial \sigma} = \sigma \Delta^2 G \quad \text{Heat Equation}$$

$$\sigma \Delta^2 G = \frac{\partial G}{\partial \sigma} = \frac{G(x, y, k\sigma) - G(x, y, \sigma)}{k\sigma - \sigma}$$

$$G(x, y, k\sigma) - G(x, y, \sigma) \approx (k-1)\sigma^2 \Delta^2 G$$

$$\text{Typical values: } \sigma = 1.6; k = \sqrt{2}$$

Building a Scale Space

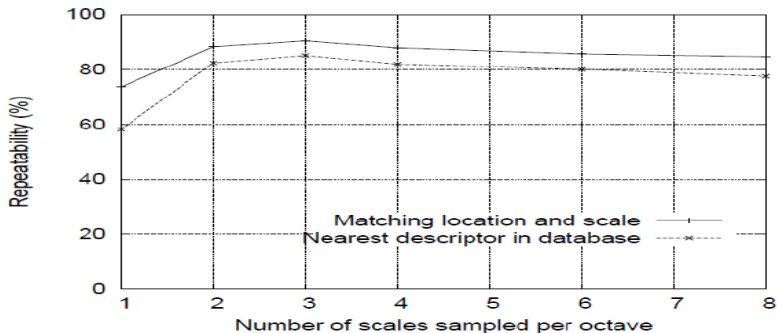


Building a Scale Space (cont'd)

	scale →				
octave	0.707107	1.000000	1.414214	2.000000	2.828427
	1.414214	2.000000	2.828427	4.000000	5.656854
	2.828427	4.000000	5.656854	8.000000	11.313708
	5.656854	8.000000	11.313708	16.000000	22.627417

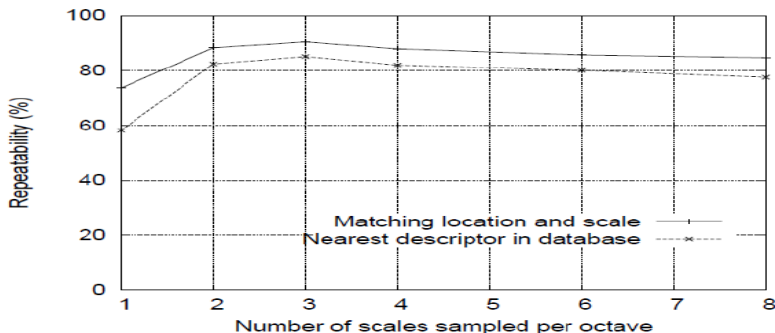
$$\sigma = .707187.6; k = \sqrt{2}$$

How Many Scales per Octave?



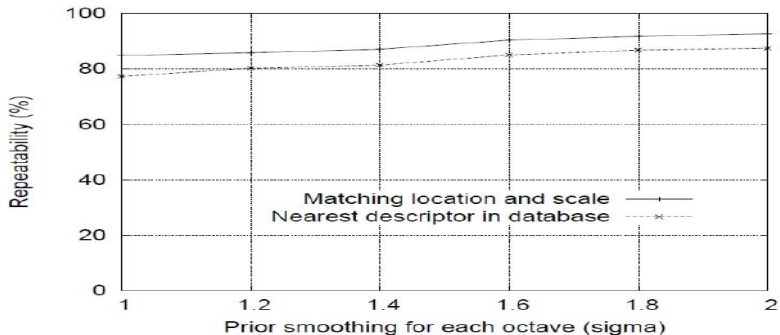
- Using 3 scales per each octave because it gives largest percentage of repeatability.

How Many Scales per Octave?



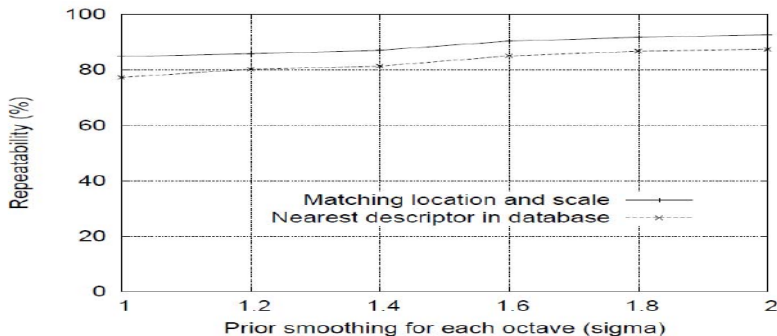
- Using **3 scales per each octave** because it gives largest percentage of repeatability.

Initial Value of Sigma



- Initial value of $\sigma=1.6$ because the percentage of repeatability is flat.

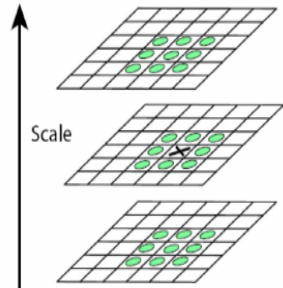
Initial Value of Sigma



- Initial value of $\sigma=1.6$ because the percentage of repeatability is flat.

Scale Space Peak Detection

- Compare a pixel (X) with 26 pixels in current and adjacent scales (Green Circles)
- Select a pixel (X) if larger/smaller than all 26 pixels
- Large number of extrema, computationally expensive
 - Detect the most stable subset with a coarse sampling of scales



Key Point Localization

- Candidates are chosen from extrema detection.



original image



extrema locations

Initial Outlier Rejection

1. Low contrast candidates
 2. Poorly localized candidates along an edge
- Taylor series expansion of DOG, $\mathbf{D}$.

$$D(\mathbf{x}) = D + \frac{\partial D}{\partial \mathbf{x}} \mathbf{x} + \frac{1}{2} \mathbf{x}^T \frac{\partial^2 D}{\partial \mathbf{x}^2} \mathbf{x} \quad \mathbf{x} = (x, y, \sigma)^T$$

- Minima or maxima is located at $\hat{\mathbf{x}} = -\frac{\partial^2 D}{\partial \mathbf{x}^2}^{-1} \frac{\partial D}{\partial \mathbf{x}}$
- Value of $D(\mathbf{x})$ at minima/maxima must be large, $|D(\mathbf{x})| > th$.

Initial Outlier Rejection (cont'd)



from 832 key points to 729 key points, $th=0.03$.



Further Outlier Rejection (cont'd)

- ▶ DoG has strong response along edge
- ▶ Assume DoG as a surface:
 - ▶ Compute principal curvatures (PC).
 - ▶ Along the edge one of the PC is very low, across the edge is high.

Initial Outlier Rejection (cont'd)

- Analogous to Harris corner detector
- Compute Hessian of D

$$\mathbf{H} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix} \quad \begin{aligned} \text{Tr}(\mathbf{H}) &= D_{xx} + D_{yy} = \lambda_1 + \lambda_2 \\ \text{Det}(\mathbf{H}) &= D_{xx}D_{yy} - (D_{xy})^2 = \lambda_1\lambda_2 \end{aligned}$$

- Remove outliers by evaluating

$$\frac{\text{Tr}(\mathbf{H})^2}{\text{Det}(\mathbf{H})} = \frac{(r+1)^2}{r} \quad r = \frac{\lambda_1}{\lambda_2}$$

Initial Outlier Rejection (cont'd)

- Following quantity is minimum when $r=1$
- It increases with r

$$\frac{Tr(H)^2}{Det(H)} = \frac{(r+1)^2}{r}$$

$$r = \frac{\lambda_1}{\lambda_2}$$

- Eliminate key points if $r > 10$

Initial Outlier Rejection (cont'd)



from 729 key points to 536 key points.

Orientation Assignment

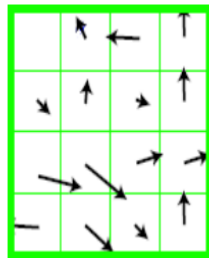
- ▶ To achieve **rotation invariance**.
- ▶ Compute **central derivatives**, **gradient magnitude** and **direction** of L (smooth image) at the scale of key point (x,y) .

$$m(x,y) = \sqrt{(L(x+1,y) - L(x-1,y))^2 + (L(x,y+1) - L(x,y-1))^2}$$

$$\theta(x,y) = \tan^{-1}((L(x,y+1) - L(x,y-1)) / (L(x+1,y) - L(x-1,y)))$$

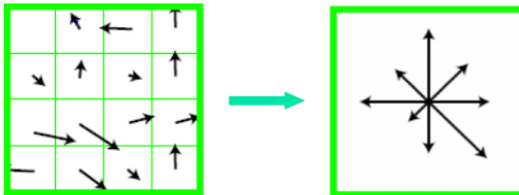
Orientation Assignment (cont'd)

- Create a weighted direction histogram in a neighborhood of a key point (36 bins)
- Weights are
 - Gradient magnitudes
 - Spatial gaussian filter with $\sigma = 1.5 \times \langle \text{scale of key point} \rangle$



Orientation Assignment (cont'd)

- Select the peak as direction of the key point
- Introduce additional key points (same location) at local peaks (within 80% of max peak) of the histogram with different directions





Local Image Descriptors at Key Points

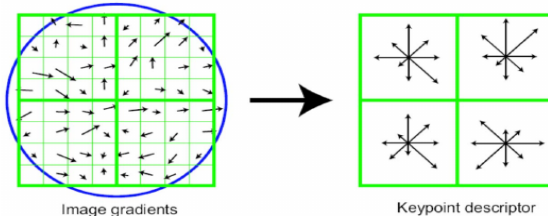
- ▶ Possible descriptor:
 - ▶ Store intensity samples in the neighborhood.
 - ▶ Sensitive to lighting changes, 3D object transformation.
- ▶ Use of gradient orientation histograms.
 - ▶ Robust representation.

Similarity to IT Cortex

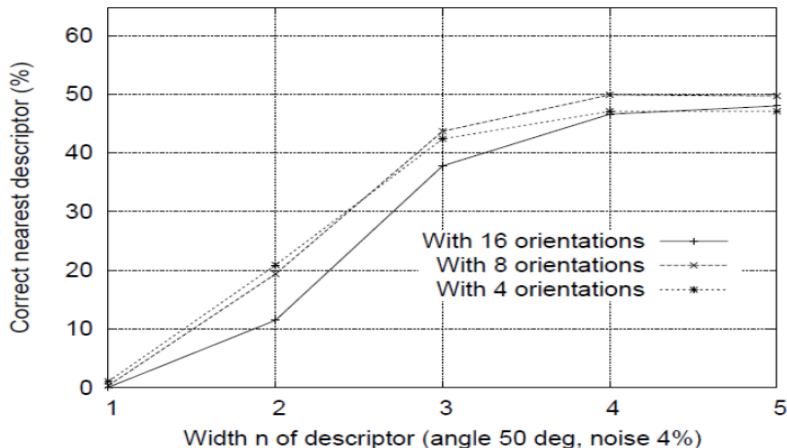
- ▶ Complex neurons **respond to a gradient** at a particular orientation.
- ▶ **Location of the feature can shift** over a small receptive field.
- ▶ Edelman, Intrator, and Poggio (1997): The function of the cells **allow for matching and recognition of 3D objects** from a range of view points.
- ▶ Experiments show better recognition accuracy for 3D objects rotated in depth by **up to 20 degrees**.

Extraction of Local Image Descriptors at Key Points

- Compute relative orientation and magnitude in a 16x16 neighborhood at key point
- Form weighted histogram (8 bin) for 4x4 regions
 - Weight by magnitude and spatial Gaussian
 - Concatenate 16 histograms in one long vector of 128 dimensions
- Example for 8x8 to 2x2 descriptors



Descriptor Regions (n by n)



Extraction of Local Image Descriptors at Key Points

- ▶ Store numbers in a vector.
- ▶ Normalize to unit vector (UN)
 - ▶ Illumination invariance (affine changes)
- ▶ For non-linear intensity transforms
 - ▶ Bound Unit Vector items to maximum 0.2 (remove larger gradients).
 - ▶ Renormalize to unit vector.



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What IS SIFT?

Steps for Extracting Key Points

Key Point Matching

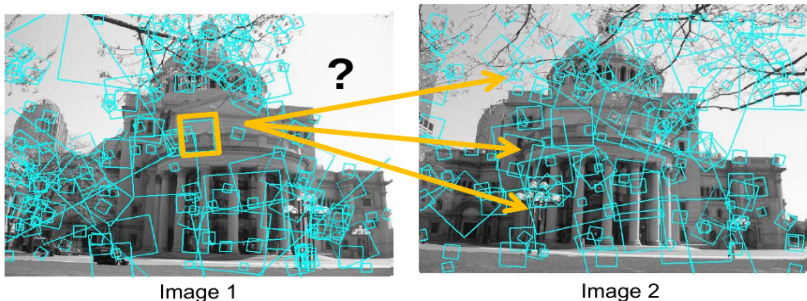
Key Point Matching

- Match the key points against a database of that obtained from training images.
- Find the nearest neighbor i.e. a key point with minimum Euclidean distance.
 - Efficient Nearest Neighbor matching
 - Looks at ratio of distance between best and 2nd best match (.8)

Matching Local Features



Matching Local Features (cont'd)



- To generate **candidate matches**, find patches that have the most similar appearance or SIFT descriptor
- Simplest approach: compare them all, take the closest (or closest k, or within a thresholded distance)

Ambiguous Matches



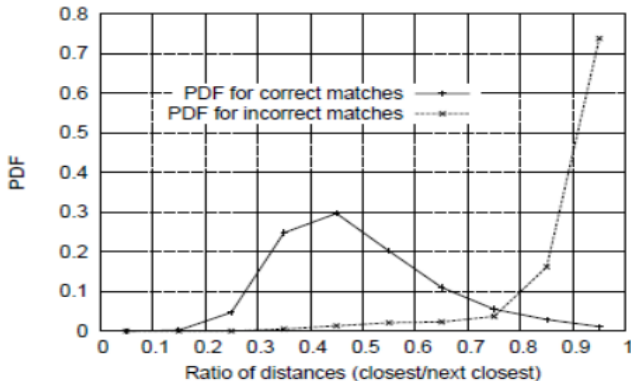
Image 1



Image 2

- At what distance do we have a good match?
- To add robustness to matching, can consider **ratio** : distance to best match / distance to second best match
- If low, first match looks good.
- If high, could be ambiguous match.

The ratio of distance from the closest to the distance of the second closest





References

1. David G Lowe: Distinctive image features from scale-invariant keypoints, International journal of computer vision, vol. 60, no. 2, pp. 91-110, 2004.
2. Mubarak Shah CV Course: **Lecture 6**
(<https://www.youtube.com/watch?v=L77m5xuDSKw&list=PLd3hISJ9&index=6>).
3. David Forsyth and Jean Ponce: Computer Vision: A Modern Approach, 2nd Ed., Pearson Education, Inc., 2012.
4. Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Ed., Springer-Verlag London, 2020.